# James Akl

**Robotics Researcher**

james-akl@outlook.com | LinkedIn | Scholar | 617-470-••••
jamesakl.com | GitHub | ORCiD | Worcester, MA


## EDUCATION Coursework

| | | | | | |
|---|---|---|---|---|---|
| **PhD** | Robotics Engineering | Worcester Polytechnic Institute | GPA 4.00/4.00 | **Aug 2019** | **May 2024** (exp.) |
| **BEng** | Mechanical Engineering | Lebanese American University | GPA 3.72/4.00 | **Sep 2015** | **May 2019** |
| **Minor** | Mathematics | Lebanese American University | GPA 4.00/4.00 | **Sep 2015** | **May 2018** |

## EXPERIENCE Details

### Doctoral Researcher — Worcester Polytechnic Institute
Manipulation & Environmental Robotics Laboratory — **Aug 2019 – PRESENT**

**Develop strategies/algorithms** for metal cutting, waste sorting, and soft robots using robot **vision**, **control**, and **machine learning**.

### Teaching Assistant — Worcester Polytechnic Institute
Robotics Engineering Department — **Aug 2019 – May 2020**

Assisted **delivery**, **labs**, and **grading** of robotics courses: *Unified Robotics I: Actuation*, *Unified Robotics II: Sensing*, *Introduction to Robotics*.

### Visiting Researcher — University of Ottawa
Mathematical Modeling & Physical Experiments — **Jun 2018 – Aug 2018**

**Modeled** the vibrations of ribbed plates, and **experimentally verified** the performance of **analytical** versus **numerical models**.

### Research Assistant — Lebanese American University
Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

**Demonstrated** the viability of the *ionic buoyancy engine* for underwater propulsion using a coupled multiphysics **numerical model**.

## SKILLS Projects

| | |
|---|---|
| **Robotics** | Kinematics & Dynamics, Control, Motion planning, Manipulation & Grasping, Robot autonomy |
| **Computing** | Algorithms, Data structures, Computer vision, Machine learning & AI, Geometric algorithms |
| **Software** | C++, Python, Bash, Git, Linux, ROS/Gazebo, MATLAB/Simulink, Mathematica, CMake |
| **Mathematics** | Linear algebra, Probability & Statistics, Analysis & Optimization, Numerical algorithms, Proofs |
| **Research** | Surveys/reviews, Mathematical modeling, Simulations, Physical experiments, Field work |
| **Publishing** | MS Office, LaTeX, Technical writing, Photo-editing, Scientific illustration/diagrams, Video-editing |
| **Social** | Presentations, Communication (technical/non-technical audiences), Collaboration (Cross-disciplinary) |
| **Language** | English (native fluency), French (professional fluency), Arabic (professional fluency) |

## RESEARCH Preprints

[2022][1] **Feature-Driven Next View Planning for Robotic Cutting in Metal Scrap Recycling** *[Under Review]*

[2021][4] **ZeroWaste Dataset: Towards Deformable Object Segmentation in Extreme Clutter** *[Under Review]*

[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation** ***IEEE CASE***

[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation** ***IEEE ICRA***

[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for [...]** ***IStructE/Elsevier Structures***

[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions** ***Elsevier MethodsX***

[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation** ***SPIE Smart Structures***

[1–4] **Superscripts indicate author position in co-authored publications.**

## AWARDS Proof

**Graduated with Distinction, 2019**
LAU School of Engineering

**LAU Most Novel Club Award, 2018**
Founded the Emerging Technologies Club

**LAU Engineering Scholarship, 2015**
Won a design and programming competition